

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



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NGHIÊN CỨU GIẢI PHÁP TỐI ƯU
CẤU TRÚC CỦA ĐỘI HÌNH ROBOT
PHỐI HỢP ĐẨY VẬT

ĐỒ ÁN TỐT NGHIỆP ĐẠI HỌC HỆ CHÍNH QUY

Ngành: Kỹ thuật Robot

HÀ NỘI - 2025

SOA and Related Works

1.1 Inertial Parameters Estimation

To the best of our knowledge, there has been no comprehensive review dedicated specifically to center-of-mass (CoM) estimation for manipulated objects. Existing surveys primarily focus on the identification of the complete set of inertial parameters rather than CoM estimation itself. A survey of Schedlinsk [1] presents a literature review of existing methodologies on how to estimate these parameters in mechanical engineering, whereas Mavrakis and Stolkin [2] is specific to robotics. According to Mavrakis and Stolkin's taxonomy, the methods for estimating inertial parameters can be categorized into three main groups by degree of interaction — purely visual, exploratory, and fixed-object.

The first visual methods relied on the assumption of a known and uniform object density in order to simplify the estimation problem by reducing it to the computation of the object's volume. In [3] the object is represented as a collection of equally sized voxels with identical masses, from which the center of mass is determined by computing the moment of inertia matrix. One of the most widely recognized approaches is presented in [4], the author decomposed a rigid body into polyhedral components with uniform density and computed the mass, and inertia tensor of each component before combining them to obtain the inertial properties of the complete object. The computation was achieved by converting the required three-dimensional volume integrals into two-dimensional surface integrals and one-dimensional line integrals using the divergence theorem, projection, and Green's theorem. However, while visual perception can accurately capture an object's shape and volume, the underlying density distribution is generally unknown, may vary throughout the object, and cannot be directly observed solely from visual information.

Exploratory and fixed-object methods, on the other hand, involve physical interaction with the object to perform the estimation task. These interactive techniques share a common reliance on measuring applied forces/wrenches and the resulting object motion—including position, velocity, and acceleration—to approximate physical laws through analytical models, often utilizing least squares optimization to solve the dynamic equations.

Tài liệu tham khảo

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- [4] Brian Mirtich (1996), “Fast and accurate computation of polyhedral mass properties”. *Journal of Graphics Tools*, tập 1, số 2, tr. 31–50, doi:10.1080/10867651.1996.10487458.